

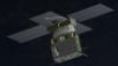


FIREFLY
A E R O S P A C E

LUKE SERRANO

Summer 26 Spacecraft GNC - Elytra

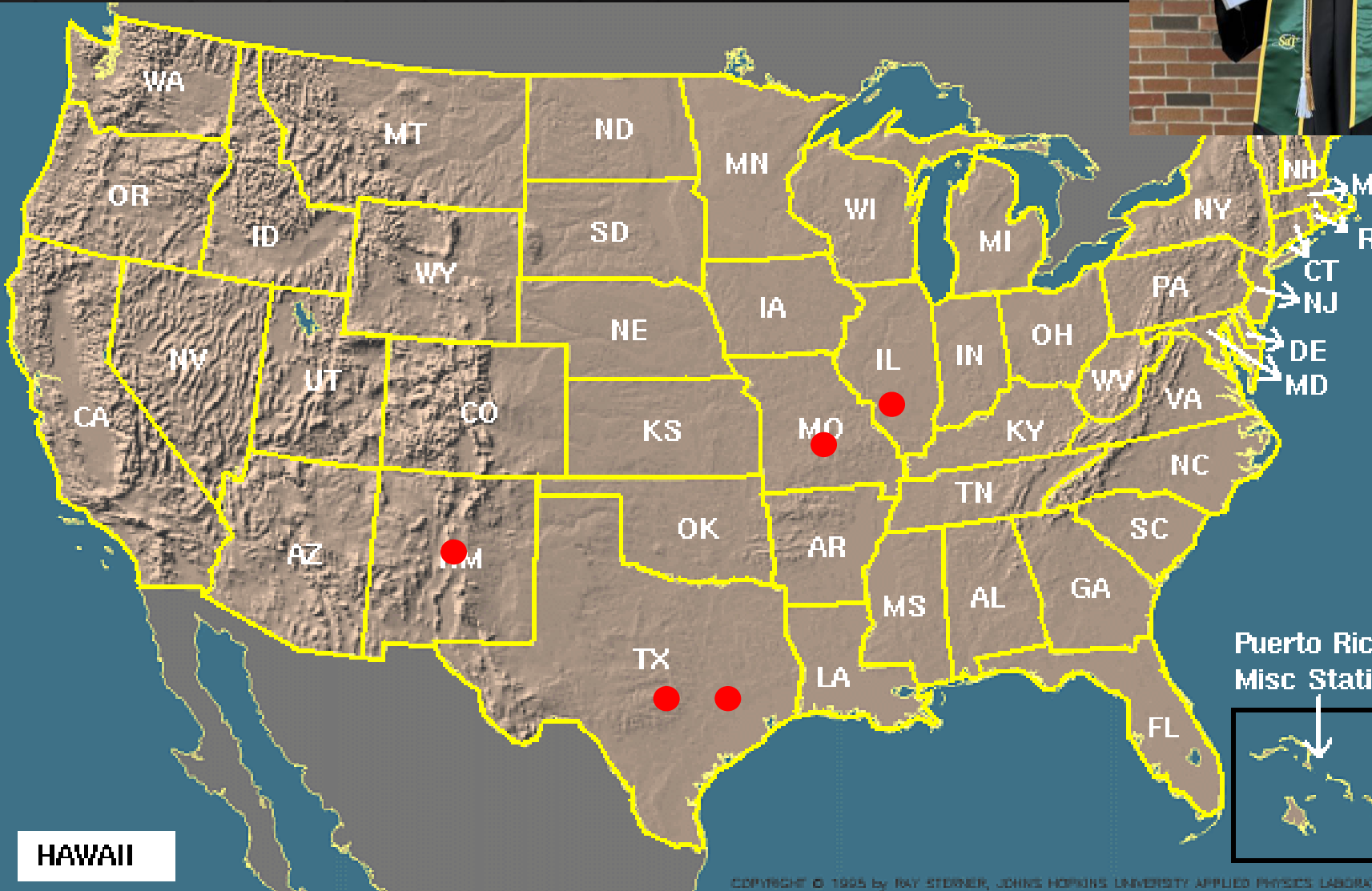
8/5/2026



- ✓ Personal Introduction
- ✓ Prior Internship Experience
- ✓ Relevant Projects / Research
- ✓ Initial Internship Tasks
- ✓ Payload Tasks
- ✓ Reflecting On This Summer / Looking Beyond This Summer
- ✓ Questions

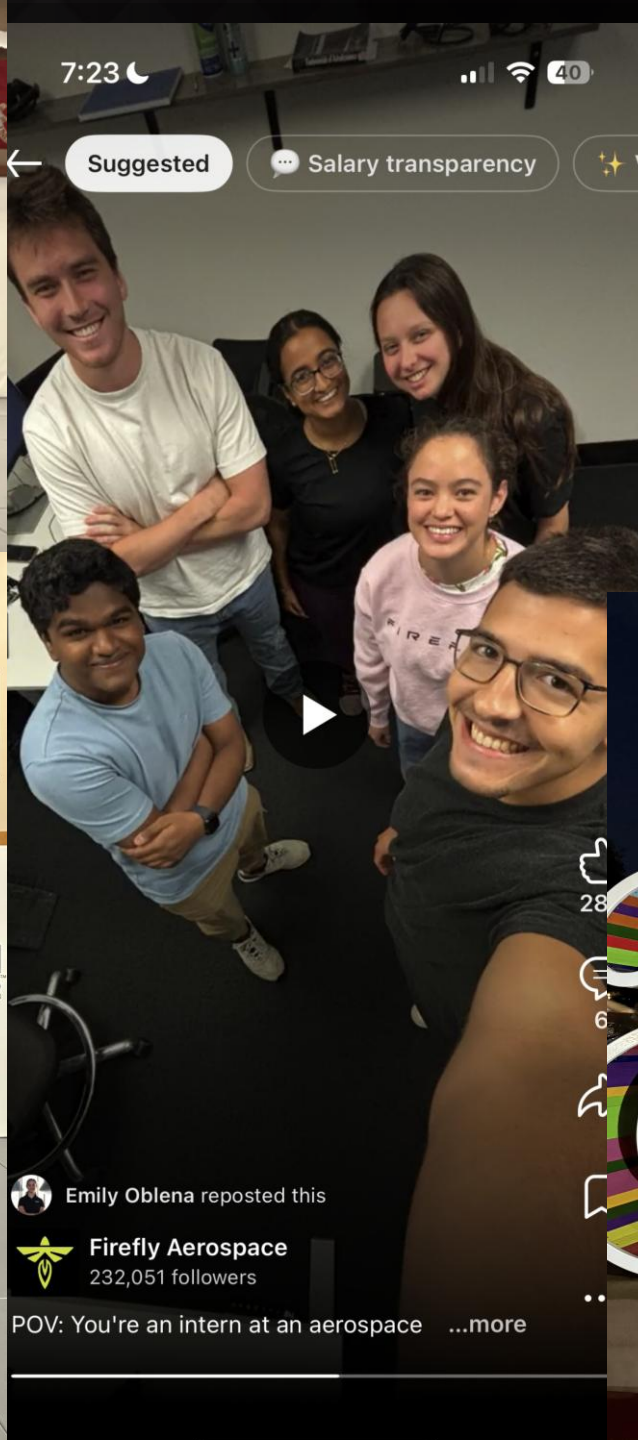
PERSONAL INTRODUCTION

B.S. Aerospace Engineering, Spring 2025
Rising 2nd year M.S. (Thesis) Aerospace Engineering



23 Years Old





PRIOR EXPERIENCE

MISSILE GNC ENGINEER (SANDIA NATIONAL LABS)

- Year-round Intern, 20 continuous months of employment, January 2024 – August 2025
- GNC and Embedded Systems team
- GNC software suites : SIL, HWIL, Pub-Sub messaging, Auto-Generated messages, FSW verification, GNC ground software

- GPS + Inertial Navigation System (INS) Navigation, cold gas ACS
- Ownership of Navigation regression test framework
- IMU calibration / tumble test anomaly
- ACS 'machine gun test' analysis and improvement



LUNAR LANDER
EKF / MEKF
FALL 2025

LAND, AIR, AND SPACE ROBOTICS LAB (LASR)

INTERNSHIP TASKS

MEASUREMENT DEGRADATION

INCREASED FIDELITY OF RSO DYNAMICS

ATTITUDE INITIALIZATION TOOL

PAYLOAD: INTRODUCTION

**PAYLOAD:
SYNTHETIC
STARFIELD IMAGE
GENERATION**

**PAYLOAD:
ANGLES ONLY
NAVIGATION (AON)
MEASUREMENT
MODELING**

PAYLOAD: FLAT SAT TESTING

REFLECTING ON THIS SUMMER

LOOKING BEYOND THIS SUMMER

QUESTIONS